



UNIVERSITÀ di VERONA

Machine Learning & Deep Learning (Deep Learning)

Cardiac Arrhythmia Detection Using Deep Learning from ECG Signals

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1. Abstract

Cardiovascular diseases continue being the reason for deaths of many people in the world; therefore, diagnosing arrhythmias in the heart is a priority task. ECG is considered the gold standard and is non-invasive; however, analysis of the results obtained by humans is a time-consuming process that involves many errors. The aim of this study is to explore the effectiveness of deep learning models in multi-class classification of heartbeats based on MIT-BIH Arrhythmia Database. In order to overcome class imbalance problem of the dataset, resampling was implemented to balance classes to 3,000 records. The signals were normalized and sequentially transformed to create 5 models: 1D CNN, Residual CNN, GRU, CNN-LSTM, and Transformer. All of them were tested using such measures as Accuracy, Precision, Recall, F1-Score, Confusion Matrices, and ROC-AUC. According to the results, the baseline 1D CNN proved to be the most efficient model with an accuracy of 92.2%, Macro F1-score 0.9219, and ROC-AUC 0.9917.

2. MOTIVATION AND RATIONALE

2.1 Context and Research Theme

Electrocardiogram (ECG) is an essential diagnostic technique that records the electrical activities of the heart. Detecting arrhythmia, which involves abnormal beating of the heart, is crucial when dealing with serious cases of heart disease, and helps prevent deadly heart diseases. In conventional techniques of analyzing ECG signals, cardiologists manually analyze the recorded ECG signals. This is very tedious and error-prone due to the large amount of data. To address this challenge, automated approaches utilizing publicly available benchmark datasets such as the MIT-BIH Arrhythmia Database [1] have become increasingly popular for arrhythmia detection research.

2.2 Problem Addressed & Significance

The main problem with the automatic multi-class arrhythmia classification lies in the development of an algorithm that can identify small differences in the morphology of the ECG signal. Additionally, real-life medical datasets are highly unbalanced. In the raw MIT-BIH database used in the experiment, the 'Normal' type of heartbeats represents 82.8%, whereas very rare anomalies, such as 'Fusion' heartbeats, account for merely 0.7%. The training process of machine learning algorithms favors the dominant class without proper preprocessing.

3. STATE OF THE ART (SOTA)

3.1 Deep Learning for Time-Series and Signal Classification

Deep learning is considered one of the most powerful technologies in biomedical signal analysis, as it enables automatic extraction of important patterns from large volumes of data. ECG signals are fundamentally time-series data, making them well suited for deep learning-based analysis.

CNN showed excellent results in classifying ECG signals because it is capable of identifying local patterns like P, QRS, and T waves. CNN convolutional filters allow recognizing peculiarities of each heartbeats category.

The task of recurrent neural networks (LSTM, GRU, etc.) is modeling sequential dependencies. Recurrent neural networks possess internal memory state; therefore, they are capable of identifying temporal dependencies between various points in ECG signals. So, recurrent architecture is appropriate for analysis of heartbeat sequences.

Recently, the Transformer architecture has gained popularity because of its ability to use self-attention mechanism. In contrast to RNN, Transformers process whole sequence at once and learn dependencies between remote points in the signal. Therefore, nowadays ECG classification researches involve analysis of CNN, RNN, CNN-RNN combination, and Transformer architectures to define the best one for arrhythmia identification.

3.2 Architecture Descriptions

3.2.1 1D CNN

1D CNN is used as the base model in this research study, where convolutional layers are applied to move filters on the time series, which helps to extract higher level features from the input ECG signal.

3.2.2 Residual CNN

In residual CNN, skip connection (i.e., adding the input of a layer to its output) is used [2]. Through this structure, a deeper network can be achieved, as this network avoids the vanishing gradient problem.

3.2.3 GRU

Gated Recurrent Units (GRUs) are processed in a sequence-wise manner [4]. They employ gating techniques in order to regulate the information flow, keeping the historical context of the ECG signal relevant to the present step without using the computational power of LSTMs.

3.2.4 CNN+LSTM

The architecture combines the benefits of both approaches. First, 1D Convolutional layers analyze spatial morphological characteristics of a heartbeat and after that feed this information into the LSTM layer [3] to predict the sequence and rhythms.

3.2.5 Transformer

The architecture was initially invented for use in natural language processing and employs the self-attention technique [5]. This enables the model to assign weights to each time step in the ECG signal with respect to all other time steps.

4. OBJECTIVES

The primary objectives of this project are:

1. To develop an automated ECG heartbeat classification framework using deep learning techniques.
2. To preprocess and balance the MIT-BIH Arrhythmia Dataset to address severe class imbalance.
3. To implement and compare multiple deep learning architectures, including 1D CNN, Residual CNN, GRU, CNN-LSTM, and Transformer models.
4. To evaluate model performance using Accuracy, Precision, Recall, F1-score, Confusion Matrix analysis, and ROC-AUC metrics.
5. To identify the most effective architecture for multi-class arrhythmia classification and analyze its strengths and limitations.
6. To investigate the applicability of deep learning methods in intelligent healthcare and computer-aided diagnosis systems.

5. METHODOLOGY

5.1 Dataset

In this work, the MIT-BIH Arrhythmia dataset has been used [1], which is one of the most popular benchmark datasets for automatic classification of heartbeats in an ECG. The dataset is annotated and contains recordings of ECG heartbeats from different patients belonging to several heartbeat categories.

Every ECG heartbeat instance has 187 feature values along with a class label specifying the category of that heartbeat. There are five heartbeat classes:

- Class 0: Normal Beat (N)
- Class 1: Supraventricular Ectopic Beat (S)
- Class 2: Ventricular Ectopic Beat (V)
- Class 3: Fusion Beat (F)
- Class 4: Unknown Beat (Q)

The original dataset exhibits severe class imbalance. Normal heartbeats dominate the dataset, while minority arrhythmia classes contain significantly fewer samples. Such an imbalance can negatively affect model performance, as learning algorithms tend to favor the majority class and overlook rare but clinically important arrhythmias.

Table 5.1 Original and Balanced Class Distribution

Class	Heartbeat Type	Raw Count	Percentage	Balanced Count
0	Normal	72,471	82.8%	3,000
1	Supraventricular	2,223	2.5%	3,000
2	Ventricular	5,788	6.6%	3,000
3	Fusion	641	0.7%	3,000
4	Unknown	6,431	7.3%	3,000

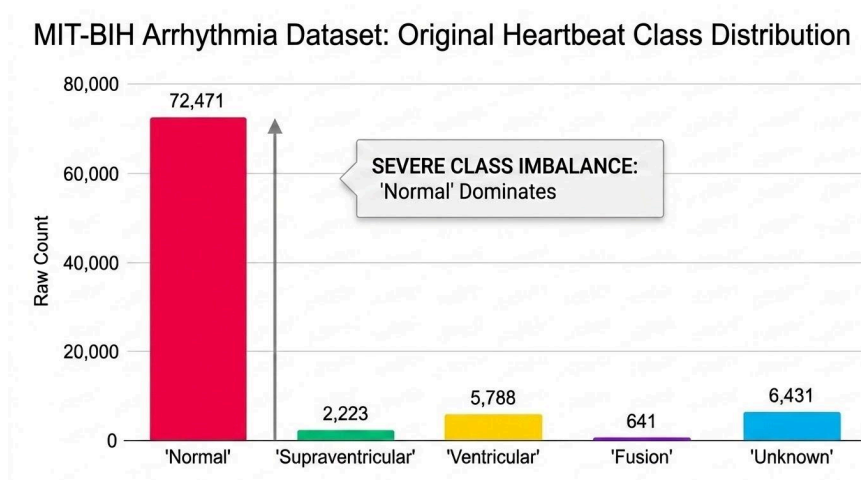


Figure 5.1: Original and balanced class distributions of the MIT-BIH Arrhythmia Dataset.

5.2 Data Preprocessing Pipeline

Data preprocessing plays a crucial role in the overall efficiency of deep learning models. Several preprocessing steps were undertaken prior to model training.

Data Loading

The ECG heartbeat signals were imported from the MIT-BIH Arrhythmia Database and split into feature vectors and class labels. Each record consists of 187 ECG signal values along with a corresponding class label.

Data Balancing

In the original dataset, there was a noticeable imbalance between heartbeat classes. To eliminate the classifier's tendency to favor the dominant class, a random resampling technique was applied. Each class was resampled to contain an equal number of samples.

Signal Normalization

The ECG signals were normalized using the Min-Max normalization technique. It transforms signal values to the [0,1] range, improving numerical stability and speeding up the training process.

The normalization formula is represented as follows:

where:

- (X) represents the original signal value,
- ($X_{\min}$) is the minimum signal value,
- ($X_{\max}$) is the maximum signal value.

$$X_{\text{normalized}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Reshaping

Since ECG data represent sequential time-series signals, the normalized samples were reshaped into the following format:

(samples, timesteps, features),

resulting in a shape of (3000, 187, 1) f

or each heartbeat class prior to the train-validation split.

Data Splitting

The balanced dataset was divided into training, validation, and testing subsets. The training data were used to optimize model parameters, the validation data were used to monitor performance during training and prevent overfitting, while the testing data were reserved exclusively for final performance evaluation. The final dataset consisted of 12,750 training samples, 2,250 validation samples, and 1,500 test samples.

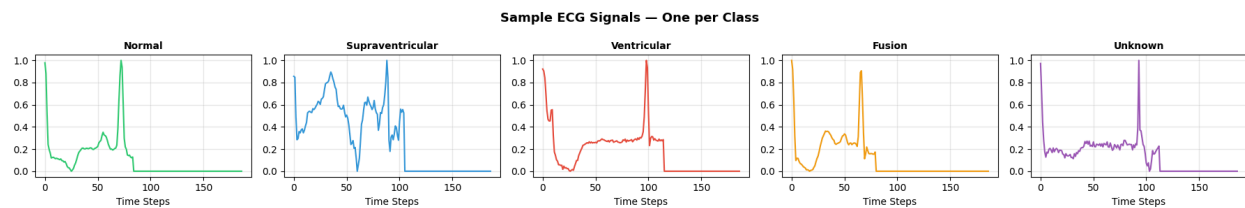


Figure 5.2: Sample ECG heartbeat signals from the five heartbeat categories.

5.3 Hyperparameter Configuration

All deep learning architectures were trained using identical hyperparameter settings to ensure a fair comparison among models.

Hyperparameter	Value
Optimizer	Adam
Learning Rate	0.001
Batch Size	64
Maximum Epochs	20

Hyperparameter	Value
Early Stopping	Patience = 7 (monitor: val_loss)
Learning Rate Reduction	Factor = 0.5, Patience = 5
Loss Function	Categorical Crossentropy
Output Activation	Softmax

Table 5.2: Hyperparameter Configuration

The Adam optimizer was selected due to its adaptive learning capability and efficient convergence behavior. Early stopping was employed to prevent overfitting by halting training once the validation loss stopped improving.

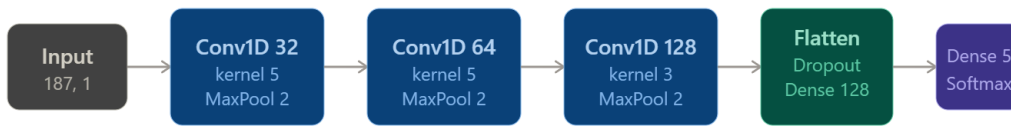
5.4 Deep Learning Architectures

Five deep learning architectures were implemented and evaluated.

5.4.1 One-Dimensional Convolutional Neural Network (1D CNN)

The 1D CNN model applies 1D convolutional layers to extract discriminative local features of the input signal. Convolutional layers can detect important morphological features such as P waves, QRS complex, and T waves, which are crucial for heartbeat classification. Pooling layers are used to decrease the dimensionality of the feature maps while keeping the most discriminative properties of the signal. Extracted features are further used in fully connected layers for multi-class classification purposes.

The developed network includes three convolutional blocks with 32, 64, and 128 filters, respectively. Each convolutional layer is followed by a max-pooling layer to decrease the dimensions of the feature maps. After the process of feature extraction, the resulting feature maps are flattened and passed to a dropout layer to prevent overfitting. Then, a dense layer with 128 neurons extracts higher-level features of the input signal. The network concludes with a softmax output layer consisting of five neurons representing the five heartbeat classes. The developed architecture will be used as a baseline model in this study, since CNNs have been successfully applied to biomedical signal processing.



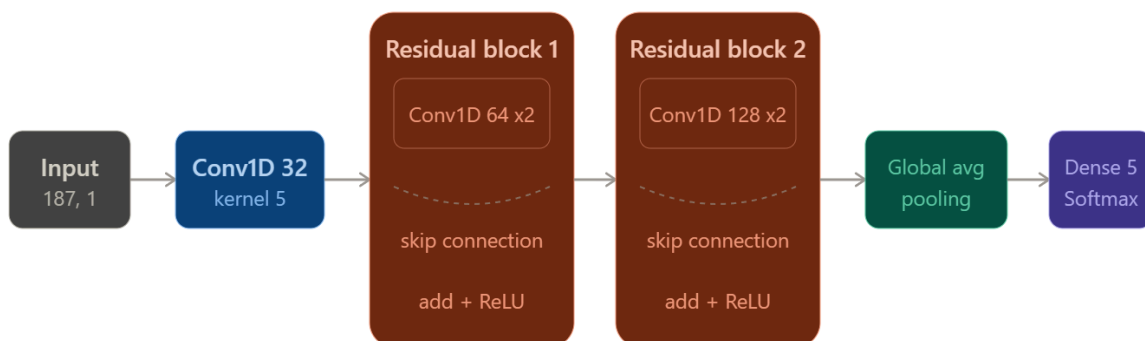
Three convolution + pooling blocks extract local waveform features before classification

Figure 5.3: 1D CNN model architecture

5.4.2 Residual Convolutional Neural Network

The Residual CNN is an enhanced version of the classical CNN model, where residual (skip) connections are added [2], allowing information to bypass intermediate layers during forward propagation. The shortcut connections contribute to solving the vanishing gradient problem and make it possible to train deeper neural networks. In other words, residual networks are capable of learning more sophisticated feature representations without compromising their optimization performance.

The developed Residual CNN contains an initial convolutional layer for the extraction of basic ECG features. This is followed by two residual blocks containing 64 and 128 filters, respectively. Each residual block contains two convolutional layers and a shortcut connection for summing the input feature map with the output of the residual block. After passing through the residual blocks, the learned feature maps are aggregated using global average pooling. Finally, a fully-connected classification layer with the softmax function estimates the probability distribution over the five heartbeat classes.



Skip connections let the input bypass the convolution layers, easing gradient flow in deeper networks

Figure 5.4: Residual CNN model architecture

5.4.3 Gated Recurrent Unit (GRU)

The Gated Recurrent Unit (GRU) model is a recurrent neural network that can learn temporal dependencies of sequential signals. GRU cells use the update and reset gates [4] to control the flow of information through the network. In comparison with traditional recurrent networks, GRUs have fewer parameters but provide strong sequential modeling performance.

In this study, the GRU layer is preceded by two convolutional layers (32 and 64 filters) that extract local features before sequential modeling. The output of the GRU layer (64 units) is passed through a dropout layer and a dense layer with 128 neurons, followed by a softmax output layer with five neurons.

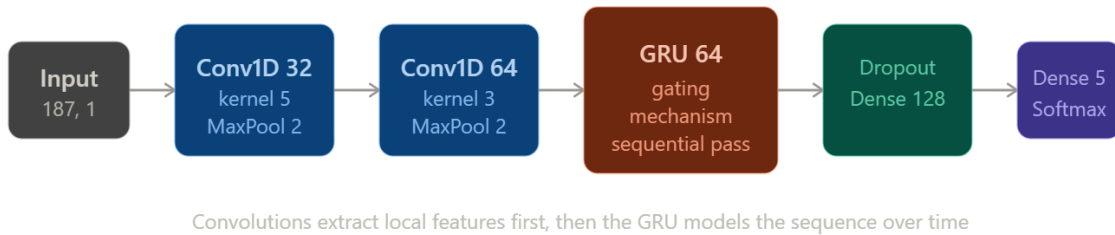


Figure 5.5: GRU model architecture

5.4.4 CNN-LSTM Hybrid Network

The CNN-LSTM model incorporates convolutional feature extraction and recurrent temporal analysis of ECG signals. At first, convolutional layers extract local features of ECG waveforms. Later, LSTM layers [3] process temporal dependencies among the features. Such hybrid architecture takes advantage of both CNN and recurrent neural networks.

The architecture consists of three convolutional blocks with 32, 64, and 128 filters, followed by an LSTM layer with 64 units. The extracted temporal features then pass through a dropout layer, a dense layer with 128 neurons, and a final softmax layer with five neurons.

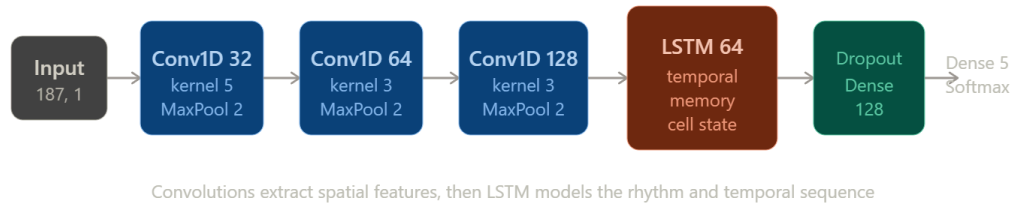


Figure 5.6: CNN+LSTM model architecture

5.4.5 Transformer Network

The Transformer model uses a self-attention mechanism [5] to capture relationships between different positions in the ECG signal. In contrast to recurrent models, transformers analyze all elements of the sequence in parallel and thus allow effective learning of both local and global dependencies. The attention mechanism allows the model to focus on the most relevant parts of the signal during classification.

In this study, the input signal is first processed by a convolutional layer before being projected into a lower-dimensional space and passed through two Transformer encoder blocks, each using two attention heads. Global average pooling is then applied, followed by a dense layer and a final softmax output layer.

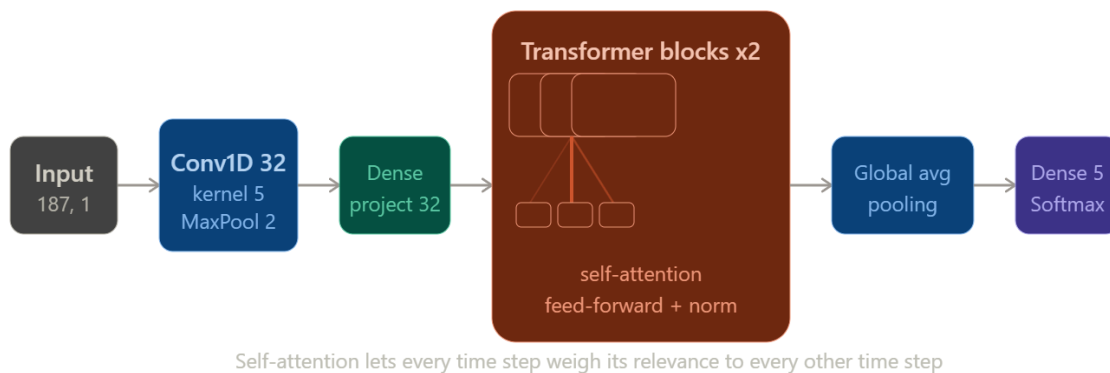


Figure 5.7: Transformer model architecture

5.5 Model Evaluation Metrics

In order to perform an all-encompassing assessment of the model's performance, several evaluation metrics were used.

Accuracy

Accuracy is the fraction of correct classifications out of all heartbeat classifications made

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

Precision is the proportion of true positives among all samples predicted as positive.

$$Precision = \frac{TP}{TP + FP}$$

Recall

Recall is the fraction of true positives out of the total number of actual positives.

$$Recall = \frac{TP}{TP + FN}$$

F1-Score

F1-score is the harmonic mean of Precision and Recall.

$$F1 = 2 * \frac{Precision \times Recall}{Precision + Recall}$$

Confusion Matrix

The confusion matrix provides a detailed overview of classification performance by displaying both correct and incorrect predictions for each heartbeat class.

ROC and AUC

ROC curves are used to evaluate the classification performance of a model at different classification thresholds. The Area Under the Curve (AUC) provides a threshold-independent measure of overall classification performance. Higher ROC-AUC values are indicative of better discrimination ability and therefore improved classification performance.

6. Experiments and Results

6.1 Training Behavior

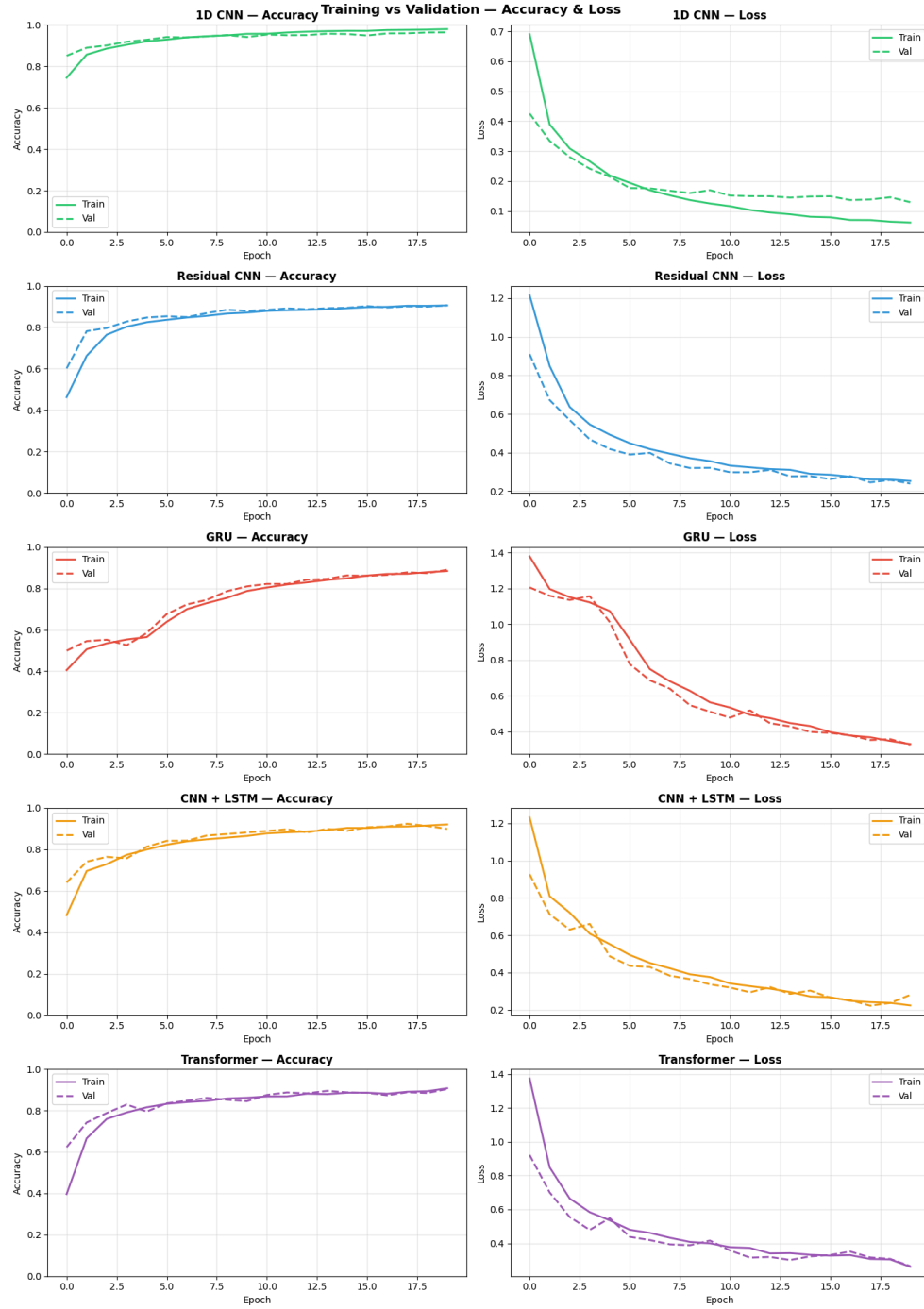


Figure 6.1: Training and validation accuracy/loss curves for all models.

Figure 6.1 shows the training and validation accuracy/loss curves for all considered models. The baseline 1D CNN model converged fastest and was the most stable, achieving high validation accuracy within a few epochs with only a small discrepancy between training and validation results. The models based on the Residual CNN and Transformer architectures exhibited some additional volatility in validation loss during the early stages of training, caused by their more complex structures. Nevertheless, with the help of the ReduceLROnPlateau callback, both models managed to stabilize and converge toward optimal values.

The GRU and CNN-LSTM models converged more slowly compared to the convolution-based architectures and showed a larger discrepancy between training and validation results, indicating a greater tendency toward overfitting. Overall, this analysis of training behavior suggests that convolution-based architectures are particularly efficient for extracting features from short ECG heartbeat signals.

6.2 Per-Model Classification Reports

1D CNN (Accuracy: 92.20%)

Class	Precision	Recall	F1-Score	Support
Normal	0.86	0.94	0.90	300
Supraventricular	0.94	0.85	0.89	300
Ventricular	0.88	0.95	0.91	300
Fusion	0.95	0.88	0.92	300
Unknown	0.99	0.99	0.99	300

Residual CNN (Accuracy: 90.40%)

Class	Precision	Recall	F1-Score	Support
Normal	0.81	0.88	0.84	300
Supraventricular	0.87	0.81	0.84	300
Ventricular	0.96	0.91	0.93	300
Fusion	0.91	0.94	0.93	300
Unknown	0.97	0.98	0.98	300

CNN+LSTM (Accuracy: 88.80%)

Class	Precision	Recall	F1-Score	Support
Normal	0.79	0.86	0.82	300
Supraventricular	0.88	0.80	0.83	300
Ventricular	0.89	0.93	0.91	300
Fusion	0.91	0.89	0.90	300
Unknown	0.98	0.97	0.98	300

Transformer (Accuracy: 88.20%)

Class	Precision	Recall	F1-Score	Support
Normal	0.84	0.75	0.79	300
Supraventricular	0.80	0.84	0.82	300
Ventricular	0.88	0.94	0.91	300
Fusion	0.91	0.91	0.91	300
Unknown	0.99	0.97	0.98	300

GRU (Accuracy: 86.40%)

Class	Precision	Recall	F1-Score	Support
Normal	0.74	0.82	0.78	300
Supraventricular	0.85	0.78	0.82	300
Ventricular	0.87	0.92	0.90	300
Fusion	0.91	0.86	0.89	300
Unknown	0.96	0.93	0.95	300

6.3 Confusion Matrices

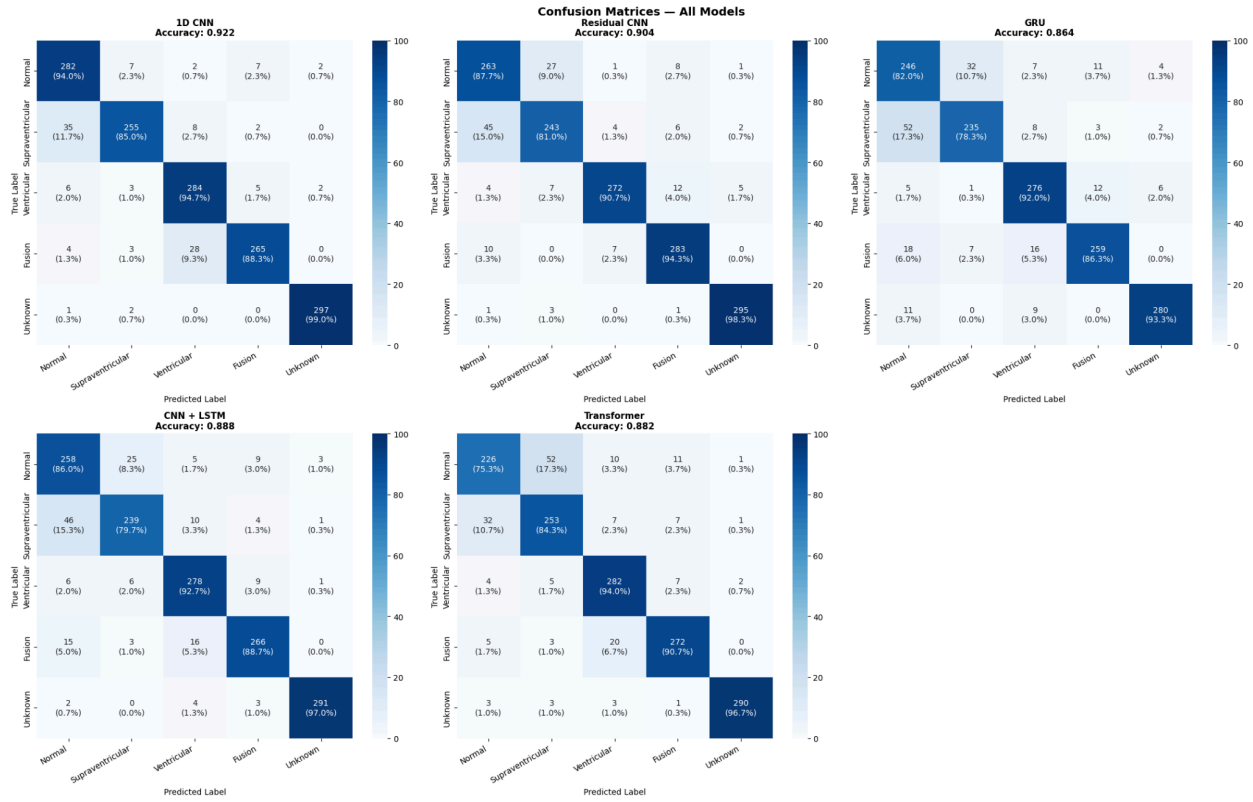
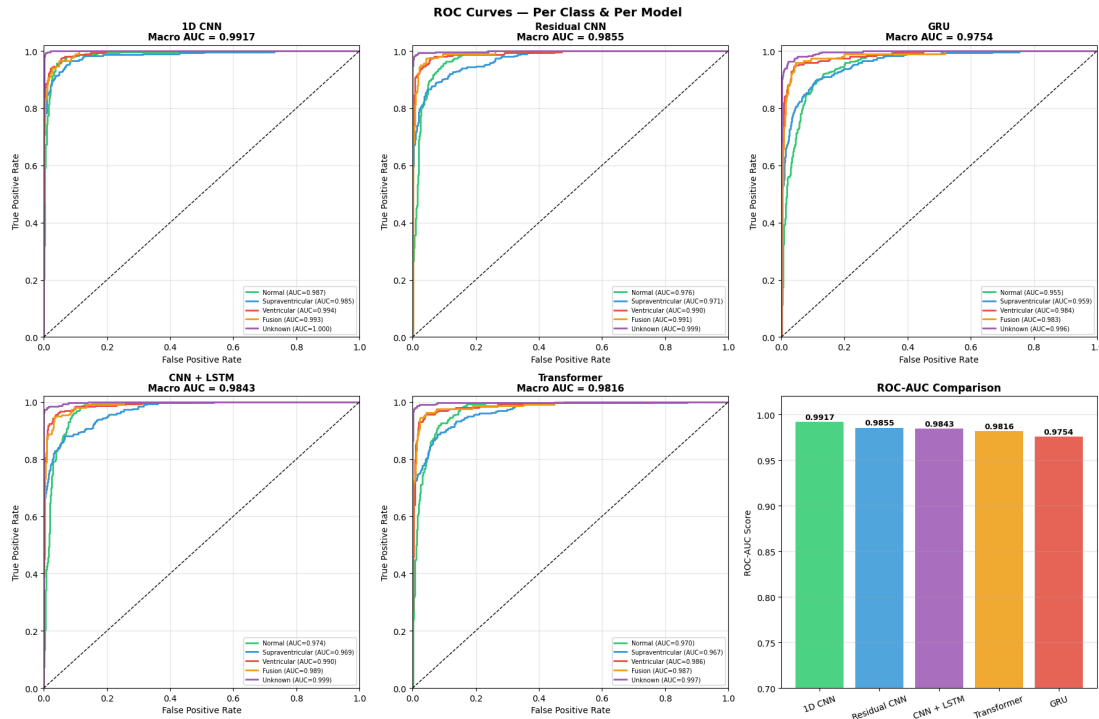


Figure 6.2: Confusion matrices for all five models.

The analysis of the confusion matrices shows that all classifiers perform well in recognizing the 'Unknown' label. On the other hand, the most significant confusion occurs between the 'Normal' and 'Supraventricular' heartbeat categories. The morphological similarity between these waveforms makes them difficult to distinguish, resulting in reduced precision and recall for both classes.

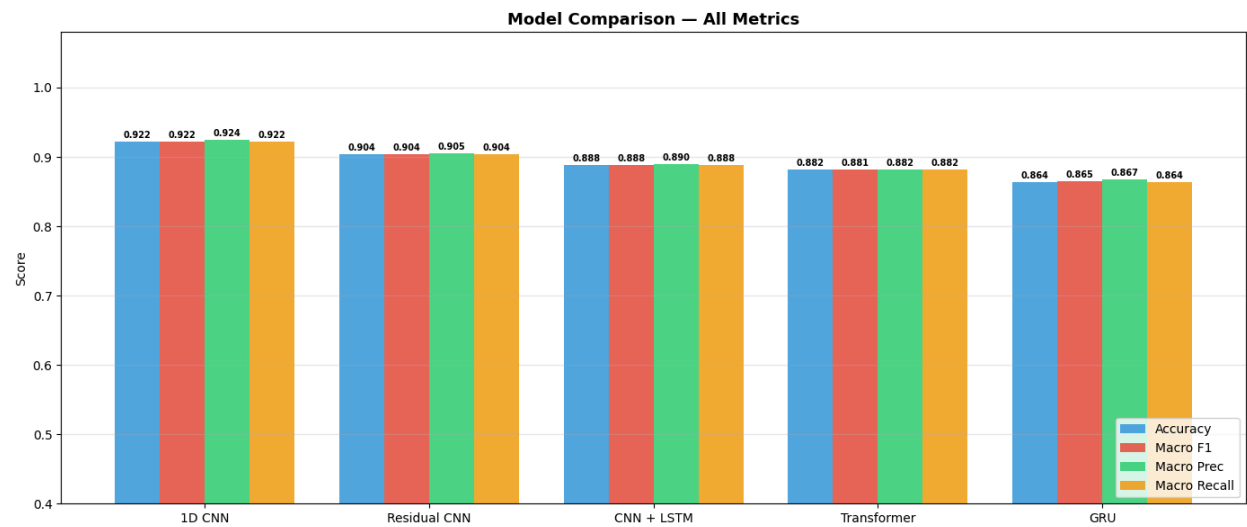
6.4 ROC-AUC Analysis



Model	ROC-AUC
1D CNN	0.9917
Residual CNN	0.9855
CNN + LSTM	0.9843
Transformer	0.9816
GRU	0.9754

To analyze the discriminative ability of each model across all heartbeat classes, Receiver Operating Characteristic (ROC) curves and Area Under the Curve (AUC) scores were employed. According to the results, the 1D CNN model achieved the best performance among all models, with a ROC-AUC value of 0.9917, indicating excellent class separation and strong classification accuracy. The Residual CNN and CNN-LSTM models also achieved strong results, with ROC-AUC values above 0.98. Although both the Transformer and GRU models showed promising results, their ROC-AUC values were slightly lower than those of the convolution-based models.

6.5 Comparative Summary



Model	Accuracy	Macro F1
1D CNN	0.922	0.922
Residual CNN	0.904	0.904
CNN+LSTM	0.888	0.888
Transformer	0.882	0.882
GRU	0.864	0.860

The comparative analysis presented in the table above shows the accuracy and Macro F1-scores of all five models, clearly indicating that convolution-based architectures consistently outperform the sequential and attention-based models. A notable performance gap exists between the best-performing model, 1D CNN (92.2%), and the weakest-performing model, GRU (86.4%). A detailed technical explanation for this performance gap is provided in the following section.

6.6 Diagnosis

According to the experimental outcomes, the 1D CNN model achieved the best classification results among all the architectures evaluated. This result can be attributed to the specific nature of ECG heartbeat data, in which local morphological features such as the P wave, QRS complex, and T wave are the most significant discriminative factors.

In addition to capturing these local characteristics, CNN filters allow for a relatively simple network design, which ensures good generalization and minimizes overfitting. Moreover, the additional complexity introduced by the GRU, CNN-LSTM, and Transformer networks brings only minor benefits, given the relatively short length of the ECG sequences.

Regarding the class-wise analysis, the 'Unknown' heartbeat category was the easiest for all models to recognize, while the greatest difficulty was observed in distinguishing between the 'Normal' and 'Supraventricular' classes due to their similar waveform shapes.

7. CONCLUSIONS

This research work presented a successful deep learning framework for automated multi-class classification of ECG arrhythmias using the MIT-BIH Arrhythmia Database. Since the dataset exhibited a significant class imbalance, a balancing approach was adopted to ensure that all classes were equally represented during the training process. Five different deep learning architectures were implemented and evaluated, including 1D CNN, Residual CNN, GRU, CNN-LSTM, and Transformer.

Among all the developed models, the 1D CNN achieved the best performance, with 92.20% accuracy, a macro F1-score of 0.922, and a ROC-AUC score of 0.9917. This research demonstrates that convolution-based architectures are particularly effective for ECG heartbeat classification, as they can readily detect local morphological features such as the P wave, QRS complex, and T wave. At the same time, more sophisticated sequential and attention-based architectures did not provide additional benefits, given the relatively short length of the ECG signal segments.

Overall, these findings suggest that local feature detection is more important than sequence modeling for this particular arrhythmia classification task. Future work could explore the incorporation of larger ECG databases, as well as the application of ensemble learning techniques.

8. References

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